

Explainable Drug-Target Affinity Prediction

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Abstract—Deep learning models excel at predicting drug-target affinity but remain frustratingly opaque, creating a serious obstacle when computational predictions need to guide real experimental work. We tackle this interpretability challenge by systematically dissecting MGraphDTA, a two-branch architecture, on the KIBA data set, using multiple complementary explainability techniques: SHAP, GNNExplainer, and Integrated Gradients. Our analysis uncovers how the model actually makes decisions: it doesn’t apply universal rules but instead learns distinct prediction strategies for different protein families, operating in a highly context-dependent manner. We show that it’s possible to extract specific, testable hypotheses from these black-box predictions by identifying particular atom-residue pairs that the model considers for its predictions, revealing its preferences for specific chemical features, and mapping which molecular substructures drive high versus low affinity predictions. This XAI framework serves to build trust in DTA predictions and transforms black-box models into collaborative tools that can generate testable hypotheses.

Index Terms—Explainable AI, Drug-Target Affinity, Graph Neural Networks, Deep Learning, Interpretability

I. INTRODUCTION

The discovery and development of novel pharmaceutical drugs is a notoriously complex, time-consuming, and capital-intensive process. An important bottleneck in this pipeline is the identification of high-affinity interactions between drug candidates (ligands) and specific protein targets. Traditional experimental methods for determining drug-target affinity (DTA), such as high-throughput screening, are accurate but often prohibitively expensive and slow to scale. Consequently, computational approaches, particularly those driven by Artificial Intelligence (AI), have emerged as indispensable tools for virtual screening, aiming to predict binding affinities with high precision before experimental validation [1], [2].

In recent years, Deep Learning has achieved state-of-the-art performance in DTA prediction. Early approaches treated molecules as simple text sequences, while more modern architectures leverage the structural richness of molecules. Notably, Graph Neural Networks (GNNs) have become the standard for representing molecular structures, allowing models to capture atom-level relationships and topological features that are lost in linear representations [3], [4]. Among these, MGraphDTA [5] represents a significant advancement, employing a dual-branch architecture that combines a DenseNet-inspired graph encoder for the ligand with a multi-scale Convolutional Neural Network (CNN) for the protein sequence.

However, despite their predictive power, these deep learning models often operate as “black boxes.” In high-stakes

domains such as pharmacology, predictive accuracy alone is insufficient. Computational chemists and biologists require transparency to trust the model’s recommendations. They must understand *why* a model predicts a high affinity: Is it reacting to a specific pharmacophore? Is it identifying a known binding pocket? Or is it exploiting data biases? While various Explainable AI (XAI) methods exist, they are rarely applied systematically to DTA models to probe both intra-molecular features and inter-molecular interactions simultaneously [6].

In this work, we address this interpretability gap. We do not propose a new predictive architecture; rather, we reproduce the MGraphDTA model on the benchmark KIBA data set and subject it to a comprehensive, multi-level explainability framework. Our contributions are as follows:

- We validate the MGraphDTA architecture, reproducing reported results on the KIBA data set with high fidelity.
- We implement a diverse toolkit of post-hoc XAI methods, including LIME [7], SHAP [7], and GNNExplainer [8], to analyze feature importance at the atomic and residue levels.
- We introduce gradient-based interaction matrices to visualize how the model “perceives” the binding mechanism between specific drug atoms and protein residues.
- We analyze the model’s latent space to understand the manifold on which chemical space is mapped.

By demystifying the decision-making process of a graph-based model, we aim to build trust in AI-driven DTA in providing testable chemical insights.

II. RELATED WORK

The computational prediction of pharmaceutical interactions has matured from a high-throughput classification task into a precise thermodynamic modeling challenge. This section traces the evolution of Drug-Target Affinity (DTA) prediction architectures, from 1D sequence models to deep geometric learning, and analyzes the trade-offs between computational efficiency and interpretability in the current state-of-the-art.

A. Problem Formulation: DTI Classification vs. DTA Regression

It is critical to distinguish between Drug-Target Interaction (DTI) and Drug-Target Affinity (DTA) prediction. DTI is formulated as a binary classification task, determining the probability $p(y = 1|d, t)$ that a drug d binds to a target t [1], [9]. While effective for filtering massive libraries, DTI

models suffer from information loss due to the arbitrary thresholding of continuous bioactivity data, for instance, treating low-affinity micromolar binders the same as high-affinity nanomolar binders [1].

In contrast, DTA prediction is a regression task targeting continuous binding constants (pK_d , pK_i , or pIC_{50}). This formulation captures the thermodynamic reality of molecular recognition, requiring models to learn a smooth manifold of chemical activity where subtle structural variations can drastically alter affinity [10].

B. The Sequence Era and the Topology Bottleneck

Early machine learning approaches relied on handcrafted descriptors and substantial domain knowledge, such as Kronecker-RLS and SimBoost [11]. The advent of deep learning shifted the paradigm towards automated feature extraction. The first wave of Deep Learning in DTA treated molecular entities as linguistic sequences. *DeepDTA* [12] served as the seminal model, employing Convolutional Neural Networks (CNNs) to process SMILES strings for drugs and FASTA sequences for proteins as 1D text. Subsequent models like *WideDTA* [13] and *DeepAffinity* [14] introduced "SMILES words" and attention mechanisms to capture broader contexts and long-range dependencies.

However, these models suffer from the "Topology Problem." A SMILES string is a linearization of a graph: atoms that are chemically bonded and spatially adjacent, for example in a ring closure, may be distant in the string representation [3]. Consequently, sequence models must expend significant capacity relearning connectivity that is explicit in geometric representations.

C. Geometric Deep Learning and MGraphDTA

To address topological infidelity, the field migrated to Graph Neural Networks (GNNs), representing molecules as graphs $G = (V, E)$. *GraphDTA* [3] demonstrated that graph encoders, such as Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) [15], outperform sequence baselines. However, it relied on "shallow" architectures (3–4 layers) due to the over-smoothing phenomenon, where deep GNNs lose expressivity as node features converge to the average [5].

MGraphDTA [5] advanced this line of work by introducing a dual-branch architecture that combines a DenseNet-inspired GNN for the drug with a multi-scale CNN for the protein sequence. The DenseNet-inspired connections enable the training of a super-deep GNN (27 layers) that avoids over-smoothing, promoting feature reuse and gradient flow. Coupled with the multi-scale CNN, this hybrid approach allows for multiscale feature extraction, simultaneously preserving high-frequency local atomic details and low-frequency global topology [5].

D. Recent paradigm: Multimodal and Transformer Architectures

The recent years offer a landscape defined by the rise of foundation models, sophisticated fusion strategies, and

attention-driven architectures. Approaches like *MixingDTA* [16] and *MolTrans* [17] leverage massive pre-trained transformers to transfer knowledge from unlabeled chemical space, achieving superior generalization in "cold-start" scenarios. Hybrid Graph-Transformer networks with dynamic prompts, as seen in HGTDPA-DTA [18], aim to capture global context better.

A critical architectural distinction exists in how interactions are modeled. Cross-Encoders, such as *MolTrans*, use expensive cross-attention mechanisms to explicitly model atom-residue pairs. This offers high interpretability but scales poorly. In contrast, Dual-Encoders, including *MGraphDTA* and *DeepDTA*, encode the drug and protein independently, fusing them only at the final stage. This is computationally efficient but lacks explicit interaction maps [19]. Furthermore, the field is seeing a rise in multimodal and structure-aware models. Approaches like MDNN-DTA [20] and MDF-DTA [21] focus on sophisticated fusion strategies, combining sequence and graph representations, while models like PocketDTA [22] explicitly incorporate 3D binding pocket information, reconciling with molecular dynamics simulations.

E. The Explainability (XAI) Gap

As model complexity grows, so does the need for Explainable AI (XAI). For general computer vision, techniques like Class Activation Mapping (CAM) and Grad-CAM [23] are standard. For tabular and text data, perturbation methods like LIME and Shapley-value-based methods like SHAP are widely used to attribute importance to input features [7].

However, explaining GNNs presents unique challenges. GNNExplainer [8] pioneered graph-native explainability by identifying compact subgraphs and feature masks that maximize mutual information with the prediction. In the context of DTA, Integrated Gradients (IG) [24] has also been employed to attribute affinity scores back to specific input tokens.

Despite these advances, a "semantic gap" remains for Dual-Encoders like MGraphDTA. While gradient methods like Grad-AAM [5] can identify *which* atoms are important, they fail to explain *why*. Specifically, they cannot reveal the specific protein residues those atoms interact with [25]. While most works focus on performance metrics, treating interpretability as a secondary visualization task, our work differentiates itself by centering the analysis on the explanation mechanisms themselves, combining graph-native methods with interaction-focused gradient analysis to visualize the "chemical reasoning" of the model.

III. MAIN IDEA

Our study is based on the MGraphDTA model, a deep learning architecture designed for drug-target binding affinity prediction [5]. The model employs a dual-encoder structure, processing drug and protein inputs through separate neural network branches before combining their representations to predict a final affinity score.

The drug, represented as a molecular graph, is processed by the ligand encoder. This encoder is a graph neural network

with an architecture inspired by DenseNet. Each atom in the drug is initialized with a 22-dimensional feature vector, being a combination of one-hot encoded categories and numerical properties as the atom type, atomic number, H-Bond Acceptor/Donor or even formal charge. The graph is then processed through a series of dense blocks composed of graph convolution layers (GraphConv). These blocks are densely connected and the input to each layer is the concatenation of the feature maps of all preceding layers within that block. A transition layer follows each dense block, and a global mean pooling operation aggregates the final node features into a single fixed-size vector representing the entire ligand.

The protein is processed by the target encoder. Protein sequences are first converted into numerical embeddings using an embedding layer. The core of this encoder is its multi-scale 1D convolutional neural network (CNN) design. It consists of multiple parallel StackCNN blocks, where each block has a different number of stacked 1D convolutional layers. This allows the model to capture patterns and motifs in the protein sequence at various scales simultaneously. Each StackCNN block is followed by an adaptive max-pooling layer to generate a fixed-size feature vector. The outputs from all parallel blocks are then concatenated and passed through a linear layer to produce the final protein representation.

Finally, the ligand and protein representation vectors are concatenated and fed into a prediction head. This head consists of a multi-layer perceptron (MLP) with several fully connected layers, interspersed with ReLU activations and dropout for regularization. The MLP processes the combined embedding vectors to regress a single, continuous value representing the predicted binding affinity.

This study focuses on the KIBA data set, where the task is to predict the KIBA score, a continuous value that quantifies the binding affinity between drugs and protein targets. The KIBA score is derived from a combination of various bioactivity measurements, including K_d , K_i , and IC_{50} values, providing a comprehensive metric for drug-target interactions [26].

IV. ALGORITHM

This section details the core explainability algorithms used to interpret the predictions of the MGraphDTA model. We employed a suite of local and global XAI techniques to provide a granular view of the specific features driving an individual affinity score, as well as to understand the model’s general behavior and biases.

A. Local Surrogate Model Explanations

We employed two popular local surrogate methods to explain individual predictions by attributing importance to human-interpretable input features.

For LIME (Local Interpretable Model-agnostic Explanations), for a given prediction, a neighborhood of perturbed samples is generated around the original input. A simple, interpretable linear model is then trained on these samples to approximate the behavior of the complex MGraphDTA model

in that local region. The weights of this linear model serve as explanations for the feature importance.

For SHAP (Shapley Additive exPlanations), we utilized the KernelSHAP variant. This method approximates the Shapley values for each feature, which represent a feature’s marginal contribution to the prediction, averaged over all possible feature coalitions. KernelSHAP explains a prediction by first sampling coalitions (subsets of features). For each coalition, two predictions are made: one with the feature of interest and one without. The difference gives the marginal contribution of that feature for that coalition. The Shapley value is then computed by taking a weighted average of these marginal contributions over all sampled coalitions. This value represents the fair contribution of the feature to the final prediction, allowing for visualizations like waterfall plots that show how features push the prediction away from a baseline value.

B. Graph-Native Explanations (GNExplainer)

To specifically probe the reasoning of the graph-based ligand encoder, we integrated GNExplainer. For a given prediction, GNExplainer identifies a minimal, critical subgraph of the original molecular graph. It does this by learning a mask over the graph’s adjacency matrix and node features, aiming to maximize the mutual information between the prediction and the distribution of possible subgraphs. The result is an intuitive explanation that highlights the most influential atoms and bonds for the model’s decision. It solves an optimization problem to learn a soft mask $\mathbf{M}$ for the graph’s adjacency matrix and a feature mask $\mathbf{F}$ that maximizes the mutual information (MI) between the model’s prediction and the masked subgraph:

$$\max_{\mathbf{M}, \mathbf{F}} \text{MI}(Y, G_S) = H(Y) - H(Y|G_S) \quad (1)$$

where Y is the model prediction and $G_S = (\mathbf{A} \odot \mathbf{M}, \mathbf{X} \odot \mathbf{F})$ is the masked subgraph. The learned masks highlight the most critical atoms, bonds, and features.

C. Gradient-Based Attribution

Gradient-based methods attribute the model’s prediction back to its input features by calculating the gradient of the output with respect to the input.

To combat the issue of vanishing gradients in saturated models, we use Integrated Gradients (IG). The attribution of an input feature x_i is calculated by integrating the gradients along a straight-line path from a baseline input x' (a zero-feature vector) to the actual input x :

$$\text{IG}_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (2)$$

where F is the model’s prediction function. The integral is approximated using a Riemann sum with $m = 150$ steps:

$$\text{IG}_i(x) \approx (x_i - x'_i) \times \frac{1}{m} \sum_{k=1}^m \frac{\partial F(x' + \frac{k}{m}(x - x'))}{\partial x_i} \quad (3)$$

Based on this, we developed two methods to generate high-resolution attribution maps. The first is a computationally efficient method where we compute the gradients of the final prediction with respect to the atom and residue feature vectors. The outer product of these two gradient vectors forms an interaction matrix, where high values indicate an important interaction.

The second method, the Atom-to-Residue Interaction Matrix, leverages IG for a more robust map. First, for a given drug-target pair, the IG attribution vector for all drug atoms, $\mathbf{A}_{\text{drug}}$, is computed. Second, the IG attribution vector for all protein residues, $\mathbf{A}_{\text{protein}}$, is computed. Finally, the interaction matrix $\mathbf{M}_{\text{int}}$, is the outer product of these two vectors:

$$\mathbf{M}_{\text{int}} = \mathbf{A}_{\text{drug}} \otimes \mathbf{A}_{\text{protein}}^T \quad (4)$$

High values in this matrix at index (i, j) indicate a strong model reliance on the interaction between drug atom i and protein residue j .

D. Global and Grouped Model Analysis

To understand the model’s general behavior and biases beyond single predictions, we used methods that analyze its internal representations and the consistency of its explanations across multiple data points.

1) *Explanation Fidelity Analysis*: To validate that the protein attention scores are causally linked to the model’s prediction, we perform a perturbation-based robustness test. This method assesses whether perturbing the most important residues (as identified by the attention mechanism) has a significantly greater impact on the model’s output than perturbing random residues.

The procedure is as follows: first, we aggregate the attention weights across all layers to obtain a single importance score for each residue in the protein sequence. From this, we identify a pool of the top 200 most important residues, which we term the “candidates.” The core of the experiment consists of randomly selecting a subset of 120 residues from this pool of 200 candidates and perturbing them. This perturbation is done either by masking the residues (i.e., replacing their feature vectors with a zero vector) or by replacing them with the features of a different amino acid from the vocabulary.

For a robust comparison, this process is repeated 10 times for each perturbation type. In parallel, a control experiment is run where, in each of the 10 repetitions, 120 residues are selected completely at random from the entire protein sequence and subjected to the same perturbation.

Finally, we measure the change in the model’s predicted affinity score for both the “top-k subset” and “random” groups. A paired t-test is then used to compare the distributions of these prediction changes. If a statistically significant larger drop in the prediction score happens when perturbing the important residues compared to the random ones, it provides strong evidence for the explanation’s fidelity. It would confirm that the attention mechanism is not focusing on arbitrary features but has successfully identified residues that are causally relevant to the model’s decision-making process.

2) *Latent Space Perturbation Analysis*: To quantify the relative importance of the information encoded in the ligand and protein embeddings, we conduct a series of latent space perturbation experiments. The analysis involves systematically zeroing out specific portions of the concatenated 192-dimensional latent vector (96 from the ligand encoder, 96 from the protein encoder) just before it is fed into the final prediction layers. We measure the resulting change in Mean Squared Error (MSE) to assess the impact. Two main experiments are performed: first, zeroing out the entire ligand or protein embedding to gauge their overall contribution; and second, zeroing out the 50% most and 50% least important dimensions, as determined by their average gradient attribution across the test set, to validate the gradient-based importance scores at a dimension-level.

3) *Context-Dependent Rule Discovery*: To investigate whether the model applies a single, universal prediction strategy or develops context-dependent “rules” for different protein targets, we analyze the relationship between the ligand embeddings and the final affinity scores. First, for each protein with a sufficient number of interaction samples ($N \geq 10$), we compute the Pearson correlation vector between its corresponding ligand embedding dimensions and the predicted affinity. This vector represents the protein’s specific “prediction rule.” We then apply hierarchical clustering to group proteins with similar rule vectors into “rule families.” Visualizing the correlation heatmap for each family allows us to assess whether the model learns distinct strategies for different protein classes, indicating context-dependent reasoning.

V. EXPERIMENTS

To demonstrate the capabilities of our multi-level explainability framework, we present in this work a comprehensive analysis of a single drug-target interaction from our test set: the binding of the protein *UniParc* UPI000004DAA4 to the protein *Uniprot ID* P36888 FLT3_HUMAN, which can be used to diagnose leukemia. For this pair, the MGraphDTA model predicts a KIBA score of 11.3629, closely matching the ground truth value of 11.400000202. The following sections dissect how the model arrived at this prediction.

We first present an independent analysis of the drug and protein components, and then investigate the interaction mechanism. While this case study focuses on a single pair for clarity, the methodology is generalizable to any drug-target pair. We conclude with a broader analysis of model behavior across multiple drugs tested against this protein target.

A. Model Validation

We used the available source code from MGraphDTA [5] to train and validate the model on the KIBA data set. Using the same hyperparameters as the original paper, including early stopping and a maximum of 3000 training epochs, we managed to reproduce the original results reported in the publication with a Mean Squared Error (MSE) of 0.1286, a concordance index (CI) of 0.9017 and a R_m^2 of 0.8021. The original results are reported in Table II.

TABLE I: Results of our reproduction of MGraphDTA on the KIBA data set compared to the original study.

Metric	Value (Ours)	Original Study [5]
MSE	0.1265	0.128
CI	0.9032	0.902
R_m^2	0.8098	0.801

TABLE II: 5-Fold Cross-Validation Results on KIBA data set with 3 repeats per fold.

Fold	Repeat	MSE	CI	R_m^2
0	0	0.15	0.89	0.77
	1	0.15	0.89	0.76
	2	0.15	0.89	0.75
1	0	0.15	0.89	0.74
	1	0.14	0.89	0.76
	2	0.14	0.89	0.76
2	0	0.14	0.89	0.77
	1	0.15	0.89	0.77
	2	0.15	0.89	0.75
3	0	0.14	0.89	0.76
	1	0.14	0.89	0.76
	2	0.14	0.89	0.74
4	0	0.14	0.89	0.76
	1	0.14	0.89	0.76
	2	0.14	0.89	0.76
Avg.		0.1448 (± 0.0030)	0.8913 (± 0.0016)	0.7575 (± 0.0086)

For further validation, we performed a more extensive evaluation using 5-fold cross-validation with 3 repeats. The detailed results for each fold and repeat are presented in Table II. As the training cost is high, we used early stopping with a maximum of 1000 epochs for this evaluation. The model achieved an average MSE of 0.1448 (± 0.0030), a CI of 0.8913 (± 0.0016), and an R^2 of 0.7575 (± 0.0086) across all folds and repeats, which are less competitive, as the maximum training epochs is only of 1000, but still comparable with the original publication. The model used for the explainability analysis described in the following sections corresponds to the model obtained from the 3000 epochs training described above.

B. Protein Analysis

We first apply KernelSHAP and LIME to understand which general, high-level residues influenced the prediction. These methods allow us to attribute specific contributions of the protein sequence input (CNN branch) to the final drug-target KIBA score.

1) *KernelSHAP Analysis*: The KernelSHAP analysis reveals that the model identifies specific residues as strong drivers for affinity prediction. As illustrated in Figure 1, the model predicts a binding affinity of 11.71, which is lower than the average base value of 12.32.

The Waterfall plot (Figure 1a) indicates that the presence of Glutamine at position 17 (Q17) acts as the primary negative contributor, depressing the output score by approximately 0.30. Secondary negative drivers include Alanine at position 393

(A393) and Lysine at position 149 (K149). The difference between the model’s prediction value of 11.36 and the final KernelSHAP output of 11.71 is explained by the approximation error inherent in the KernelSHAP method.

2) *Comparative Analysis with LIME*: We contrasted the game-theoretic approach of KernelSHAP with the local surrogate approach of LIME. As shown in Figure 2, LIME identifies Histidine at position 178 (H178) and Glutamine at position 384 (Q384) as the most significant features for local decision boundaries.

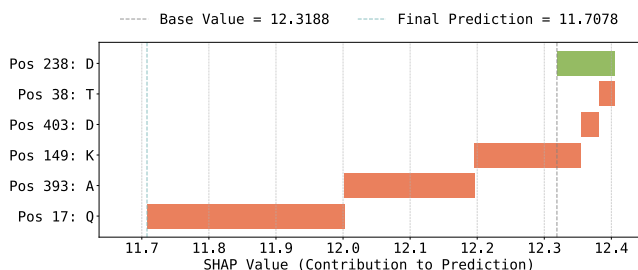
The divergence between KernelSHAP (ranking Q17 highest) and LIME (ranking H178 highest) is notable. While both methods rely on input perturbation, KernelSHAP utilizes the Shapley kernel to guarantee equitable credit assignment for feature interactions. The discrepancy suggests that the negative impact of Q17 relies on specific structural contexts captured by the Shapley constraints, whereas LIME’s local linear approximation highlights residues with high sensitivity to immediate perturbation (H178).

3) *Attention Mechanism and Explanation Fidelity*: To validate that the model focuses on distinct structural motifs, we visualized the mean attention weights (Figure 3). The heatmap demonstrates sparse, high-intensity activation bands (notably around index 0-50). Then, to further validate that the attention scores are causally linked to the model’s prediction, we performed a perturbation-based robustness test. This experiment measures the impact on the model’s output when important input features, as identified by the attention mechanism, are masked or replaced by other residues.

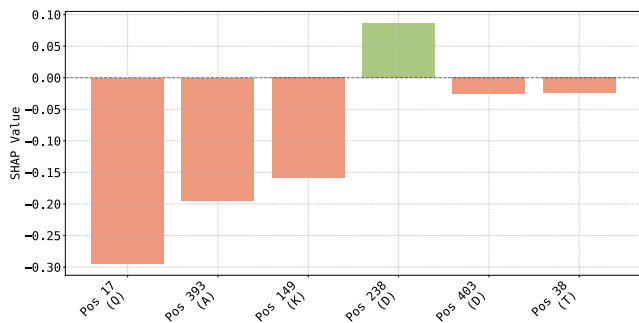
The test results, summarized in Figure 4, are nuanced. The masking experiment (“-” on the x-axis), where residues are replaced with zero vectors, shows a significantly larger drop in predicted affinity for random residues compared to masking the top-k important positions. Then, the effect of replacing residues with other amino acids is nuanced. For the first residues of the figure, introducing these residues into random locations causes a much greater impact than introducing them into the top-k important regions. This suggests the model has a low tolerance for encountering these specific residues in unexpected contexts. Conversely, for other amino acids (e.g., M, K, N), perturbing the top-k residues still results in a greater or comparable prediction drop, upholding the validity of the attention scores in those contexts. These results indicate that the attention scores are not a monolithic measure of importance. While they successfully identify residues critical to the structural integrity of the prediction, the model’s response to substitutions is highly context-dependent. It appears to have learned a non-linear, holistic representation where the impact of a perturbation depends on both the features of the new residue and the baseline expectation for the region being perturbed.

C. Ligand Analysis

To gain a more granular understanding of the ligand’s role, we use GNNExplainer at three different levels: feature, node, and edge importance. Feature importance highlights which



(a) SHAP Waterfall plot of residues and their positions.



(b) SHAP bar plot of top contributing residues.

Fig. 1: Dual visualization of KernelSHAP values for the tested protein-ligand pair.

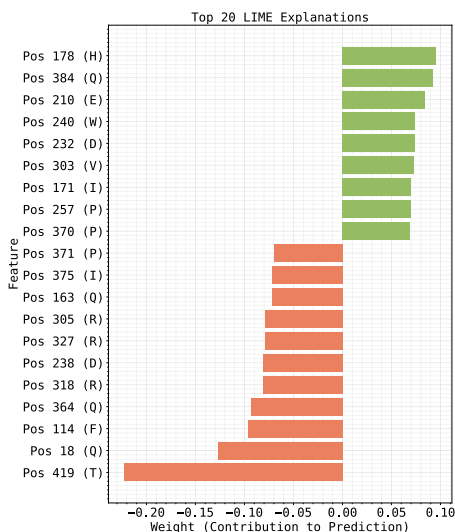


Fig. 2: LIME explanation for the protein sequence branch. Unlike KernelSHAP, LIME highlights H178 and Q384 as the most weighted features for local perturbation.

atomic features are most critical for the prediction, while node and edge importance further localize the explanation to specific atoms and bonds within the ligand structure.

Figure 5 presents the attribution scores for node features across the atoms identified as most influential. We observe that explicit Atomic Number serves as the dominant discriminatory feature across all nodes, indicating that the model prioritizes elemental identity over purely topological distinctiveness.

Complementing this abstract feature analysis, Figure 6 visualizes the node and edge importance directly on the 2D molecular structure of the ligand. It shows that the GNN branch of the DTA model focuses predominantly on chemically meaningful substructures. The highlighted regions form continuous motifs around aromatic and heteroaromatic rings, amide linkages, and electron-rich heteroatoms, indicating that the model captures local chemical context rather than relying on isolated atom features. Peripheral aliphatic substituents exhibit comparatively low importance, suggesting that the

model correctly attributes the majority of predictive signal to the core scaffold typically responsible for binding affinity.

D. Mechanism Analysis: Drug-Target Interaction

The ultimate goal of local explanation is to understand the binding mechanism. By computing the outer product of Integrated Gradients attributions for both the drug and protein, we generate an atom-residue interaction matrix, visualized in the Appendix Fig. 12 due to space constraints. Table III lists the top 5 interactions identified from this matrix. Notably, the isoleucine residue at position 181 (I181) emerges as an interaction partner, engaging with multiple drug atoms (C12, C14, C35). This suggests that I181 plays a role in stabilizing the drug-protein complex. Additionally, tryptophan at position 300 (W300) and threonine at position 419 (T419) also contribute to the predicted KIBA score.

TABLE III: Top 5 most impactful atom-residue interactions for the tested drug-protein pair as identified by the Integrated Gradients interaction explainer.

Drug Atom	Residue	Residue Position	Attribution
C14	Isoleucine	181	0.01997
C35	Isoleucine	181	-0.01917
C14	Tryptophan	300	0.01718
C14	Threonine	419	0.01716
C12	Isoleucine	181	-0.01680

To complement this tabular view, we developed a visualization that maps specific drug atoms to specific protein residues for a single pair, shown in Figure 7. The diagram displays the ligand’s 2D structure alongside a representation of the protein, with arrows connecting the most influential atom-residue pairs. This visualization is also based on Integrated Gradients, ensuring consistency with the prior interaction matrix analysis. Additional visualizations for other drugs are provided in the Appendix Section B.

Moving from a single-pair analysis to a cohort-level comparison, we next sought to determine if the model had learned to associate specific chemical motifs with binding strength. For each ligand within the top-20 high- and top-20 low-binding cohorts against the target protein, we applied GNNExplainer

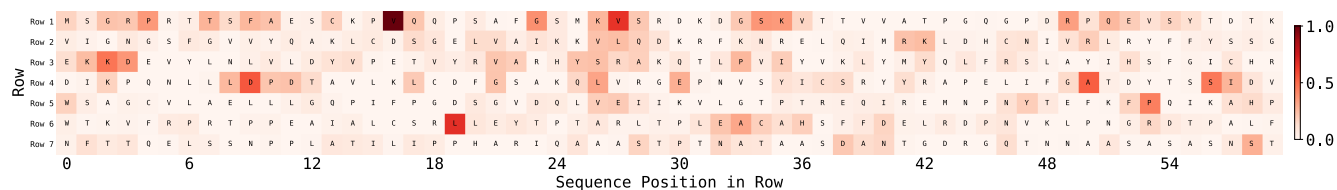


Fig. 3: Mean attention heatmap overlaying the protein sequence. Darker colors indicate regions where the model assigns higher weights, aligning with the key residues identified in the SHAP analysis. Attention scores are averaged across all attention layers and normalized for visualization. The protein sequence starts at the top left and ends at the bottom right with 60 residues per row.

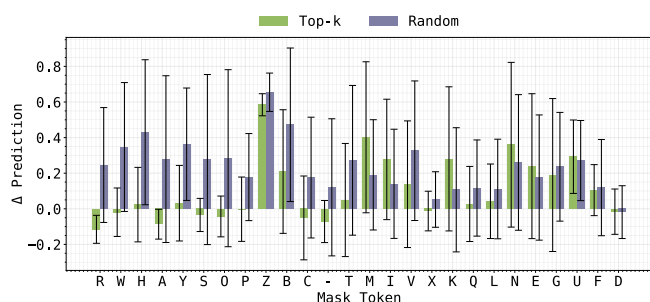


Fig. 4: Robustness analysis of protein attention scores. 120 residues out of the top 200 most important residues (as ranked by attention scores) are either masked (replaced by zero vectors, for “-” on the x-axis) or replaced by another amino acid. The change in predicted KIBA score is compared against 120 randomly selected residues.

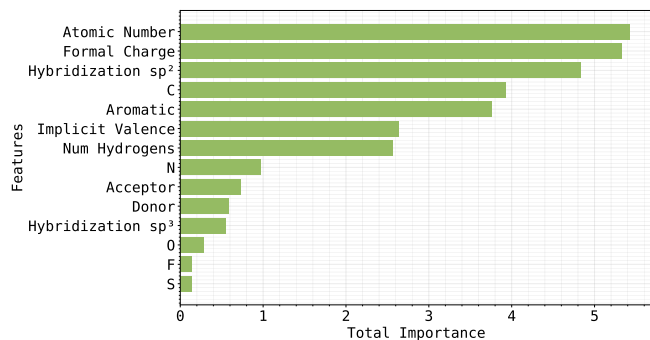


Fig. 5: GNNExplainer feature importance result for the molecular graph of the ligand. Only the 14 most important features out of the 22 atom features are shown for clarity.

to identify the top-3 most influential atoms. We then generated localized chemical descriptors for these atoms using a radius-1 Morgan Fingerprint (ECFP2) and aggregated their frequencies.

A quantitative comparison of feature importance (Table IV) reveals that at a coarse grain, the model heavily relies on aromaticity, evidenced by the high frequency of generic aromatic carbons (c) in both high- and low-binding ligands. This suggests the model correctly identifies the hydrophobic aromatic scaffold as a prerequisite for insertion into the

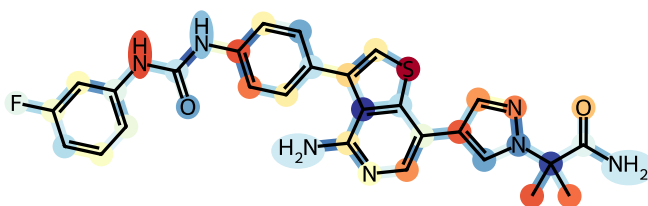


Fig. 6: GNNExplainer graph explanation highlighting important atoms and bonds in the ligand structure. Importance is indicated by color, with warmer colors denoting higher importance.

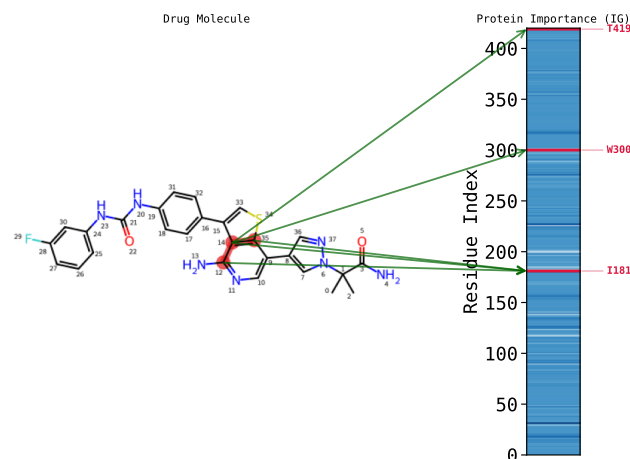


Fig. 7: Detailed interaction diagram the tested drug-target pair. Arrows indicate the top 5 predicted interactions between specific drug atoms and protein residues as shown is the top-5 table. The protein bar's color intensity reflects the attribution score for each residue.

binding pocket.

However, the distinction between high- and low-affinity ligands emerges in the presence of specific polar heterocycles. The motif N-NH-Ar, corresponding to protonated nitrogen-containing heterocycles, appears exclusively in the high-binding cohort. This indicates that the model has learned to prioritize specific Hydrogen Bond Donor (HBD) capabilities required for high-affinity interactions, likely targeting specific

residues within the protein active site. Conversely, low-binding ligands share the aromatic scaffold but often lack these specific substituted heterocyclic motifs or possess generic pyrrolic nitrogens ([nH]) that do not correlate as strongly with high affinity in this specific target context.

E. Embeddings Analysis

To assess the relative importance of the information encoded in the ligand and protein embeddings, we conducted a series of perturbation experiments. We systematically zeroed-out portions of the concatenated latent vectors before the final prediction layer and measured the impact on the model’s Mean Squared Error (MSE). The KIBA scores were binned quantile-wise into 5 groups to analyze performance across the full affinity spectrum.

First, we evaluated the impact of completely removing either the protein or ligand embedding. Figure 8 shows the MSE across different affinity bins. The results demonstrate that removing either embedding significantly degrades performance. Zeroing out the ligand embedding consistently results in a higher MSE than zeroing out the protein embedding, suggesting the model may rely marginally more on the ligand’s graph representation. This is expected as multiple ligands can bind to the same protein, making the ligand information more discriminative for affinity prediction. It should be noted that the MSE for the highest KIBA bin is equivalent when zeroing out either embedding, indicating equal contribution.

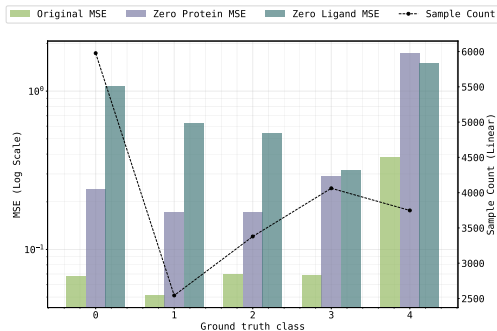


Fig. 8: Impact on MSE of zeroing out entire protein vs. ligand embeddings. The MSE axis on the left is log-scaled for clarity. The dashed line shows the sample count per bin, with its axis on the right. The Ground Truth Class represents classes of the quantile-binned KIBA scores.

Next, we performed a more fine-grained analysis to validate our gradient-based feature importance scores for the latent dimensions. We identified the 48 (50%) most and least important dimensions for both the protein and ligand embeddings based on their average gradient attribution across the test set. We then measured the MSE when zeroing out these specific dimensions. As shown in Table V, for both protein and ligand embeddings, zeroing out the most important dimensions caused a drastic increase in MSE, nearly matching the error from zeroing out the entire embedding. Conversely, zeroing

out the least important dimensions had a negligible effect on performance, validating the fidelity of our gradient-based importance scores.

F. Latent Space Analysis

We investigated the model’s latent space to determine whether it learns a single, universal set of rules for affinity prediction or if it develops context-dependent strategies.

We computed the Pearson correlation between each of the 96 dimensions of the ligand embeddings and the final affinity prediction, grouped by each unique protein sequence of the KIBA test set. Only proteins with at least 10 tested drugs were considered. Hierarchical clustering on these correlation vectors suggested the presence of three primary “rule families.” We selected three clusters as this level of granularity provided a balance between capturing distinct rule families and maintaining cluster stability. While two clusters merged distinct groups, increasing to four or more clusters resulted in the formation of very small, potentially spurious clusters, suggesting over-segmentation. Figure 9 shows the average correlation vector, or “rule,” for each of the three families. The heatmap reveals markedly different correlation profiles, suggesting the model applies different logic for each family. Cluster 0 exhibits generally weak correlations, except for a few dimensions with high positive or negative correlations (e.g., dimensions 11, 23, 43). Being the larger cluster (160 proteins), it likely represents a default strategy for general protein targets. Cluster 1 (26 proteins) shows similar patterns to cluster 0, but with an overall stronger magnitude of correlations, indicating a more decisive application of the learned rules. The last cluster is the smallest (7 proteins) and displays strong magnitude correlations across many dimensions, suggesting an outlier strategy for a unique subset of proteins.

This difference is further highlighted in Figure 10, which compares the average correlation for a subset of embedding dimensions that exhibit high variance across the identified families. For instance, dimension 35 shows a positive correlation in cluster 0, but a negative correlation in cluster 1, indicating that the same latent feature can have opposite implications depending on the protein context.

To validate that the discovered rule families were statistically significant, we conducted a permutation test. The purpose of this test was to reject the null hypothesis that the observed clusters are random groupings with no meaningful difference in their underlying “rules.” As our test statistic, we used the mean of the standard deviations of the cluster-averaged correlation vectors, a metric that quantifies the overall dissimilarity between the rules. A higher value indicates more distinct rules. We first calculated this metric for our empirically discovered clusters to get an “observed” value. Then, we generated a null distribution by running 1000 simulations. In each simulation, we randomly permuted the cluster labels among the proteins while keeping the size of each cluster fixed, and recomputed the test statistic. The p-value was calculated as the fraction of random metrics that exceeded our observed metric. As shown in Figure 11, the observed metric was a clear outlier, far

TABLE IV: Structural motifs identified by the GNN Explainer. The model distinguishes high-affinity ligands by the presence of specific protonated diazo rings and alkyl-aromatic packing, while generic pyrrolic nitrogens associate with lower affinity. Complementary visualizations are provided in the Appendix Figure 14.

Structural Motif	High	Low	Learned Association
<i>Positive Discriminators (High Affinity Drivers)</i>			
Protonated Diazo (N-NH-Ar)	3	0	Key H-Bond Donor (Specific)
Alkyl-Subst. Aromatic (Ar-R)	3	0	Hydrophobic Packing / Steric Fit
Prot. Heterocycle Link	2	0	Directed Polar Interaction
<i>Negative / Weak Discriminators</i>			
Protonated Aromatic N ([nH])	1	2	Non-specific / Suboptimal Polarity
<i>General Scaffold Features</i>			
Aromatic Carbon (c)	16	8	General Hydrophobic Scaffold
Aromatic Chain (c:c:c)	4	3	Extended Conjugation

TABLE V: MSE Results: Bold indicates lowest error within specific perturbation group (Protein or Ligand). “Half” refers to zeroing out 50% of embedding dimensions, “Least” and “Most” refer to zeroing out the 48 dimensions with the lowest and highest average gradient importance, respectively.

Class	Original	P (Half)	P (Least)	P (Most)	L (Half)	L (Least)	L (Most)
0	0.0680	0.2389	0.0862	0.1478	1.0670	0.0729	1.1055
1	0.0512	0.1715	0.0744	0.1073	0.6236	0.0513	0.5677
2	0.0696	0.1709	0.0941	0.1074	0.5419	0.0822	0.4755
3	0.0686	0.2894	0.1014	0.1192	0.3151	0.0923	0.3065
4	0.3848	1.7368	0.5414	1.2181	1.4978	0.6043	1.0542

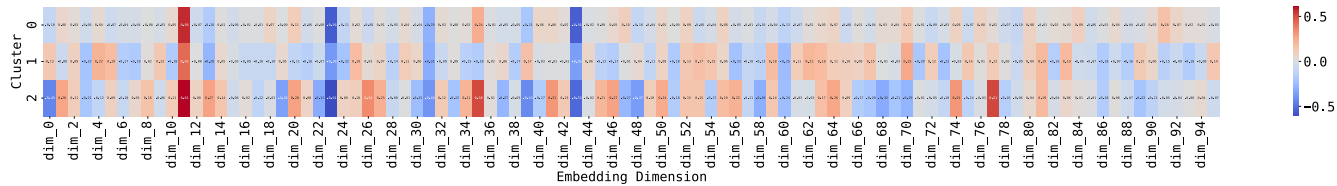


Fig. 9: Average correlation “rule” for each of the 3 identified protein families. Each row represents the average Pearson correlation between each ligand embedding dimension and the predicted KIBA score, computed across all proteins within that family. The distinct patterns across clusters indicate that the model employs different latent strategies for different protein groups.

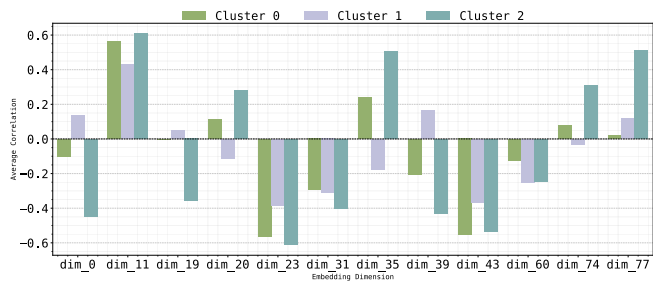


Fig. 10: Comparison of key “rules” across the three families for selected ligand embedding dimensions.

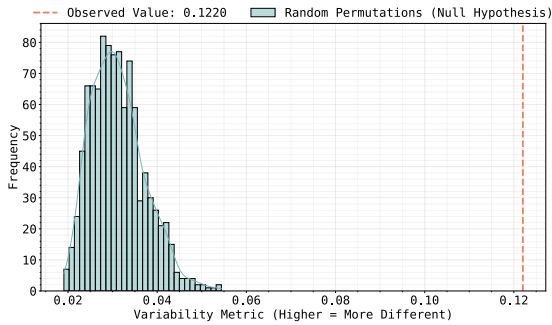


Fig. 11: Permutation test for cluster significance.

greater than any value from the random permutations, yielding a p-value < 0.001 . This result provides evidence that the discovered rule families are not a random artifact but a learned strategy of the model.

VI. DISCUSSION

A. Interpretation of Findings

Our multi-modal XAI framework reveals that the MGraphDTA model learns chemically relevant patterns, rather than relying on superficial heuristics. The GNNExplainer analysis shows the model focuses on functional "warheads" of the ligand which are known to be important for binding. This indicates that the model is not just memorizing training data, but is learning to recognize pharmacophores.

The analysis of the latent space shows the importance of ligand embeddings, which is expected as the data set contains more drug diversity than protein diversity. The discovery of "rule families" based on protein clusters suggests that the model's reasoning is context-dependent and that it applies different sets of rules for different protein families. This implies that the model may not be learning a universal, transferable model of drug-target interaction, but rather a set of specialized models for different protein classes.

The interaction explanation, based on Integrated Gradients, successfully generates specific and testable hypotheses about binding mechanisms. The identification of specific atom-residue pairs that the model considers important, such as the interaction between drug atoms and Isoleucine 181, could be a valuable starting point for medicinal chemists to guide lead optimization.

B. Comparison of XAI Methods

Each XAI method provided a different and valuable perspective on the model's behavior. GNNExplainer offers a high-level, intuitive view of important molecular substructures, making it useful for assessing what parts of a molecule the model is looking at.

For generating specific hypotheses about binding mechanisms, the granularity of Integrated Gradients proved to be more insightful. While computationally more expensive, the resulting atom-residue interaction matrices provide a much more detailed picture of the predicted binding mode.

LIME and SHAP were useful for a high-level overview of feature importance, especially for the protein sequence. However, we observed discrepancies between their outputs, with LIME highlighting different residues than KernelSHAP. This underscores the importance of not relying on a single XAI method, but rather using a suite of tools to get a more robust understanding of the model.

C. The Interactive Exploration Tool

To facilitate deeper and more accessible exploration of these findings, we developed an interactive web application using Streamlit. The application allows researchers to input novel drug and protein pairs and generate all the aforementioned explanations in real time. The source code is available at <https://github.com/413x4ndre/Drug-Target-Affinity-XAI>, and a demo application is available at <https://huggingface.co/spaces/413x4ndre/drug-target-affinity-xai>.

D. Limitations and Future Work

It is important to acknowledge the limitations of this work. First, the explanations are correlational, not causal. They show what the model is paying attention to, but they do not prove that these features are causally responsible for the predicted KIBA score.

Second, our analysis is performed on a single model architecture (MGraphDTA) and a single data set (KIBA). The findings about latent space bias and context-dependent rules may not generalize to other DTA models or data sets.

Future work should focus on prospectively validating the generated hypotheses. This could be done through collaboration with medicinal chemists, or through more computationally intensive methods like molecular dynamics simulations. Another promising direction is to use the insights gained from this XAI analysis to inform the design of more robust and interpretable model architectures. Notably, future works could integrate 3D structural information of protein-ligand complexes into the XAI framework as well as know binding sites to enhance the interpretability of spatial interactions.

Finally, the XAI framework developed in this paper could be applied to other DTA models to compare their decision-making processes.

VII. CONCLUSION

The increasing sophistication of deep learning models for Drug-Target Affinity (DTA) prediction has created an urgent need for robust interpretability methods. In this paper, we addressed the "black-box" problem by applying a comprehensive, multi-modal XAI framework to the MGraphDTA model. Our analysis successfully moved beyond simple accuracy metrics to dissect the model's internal decision-making processes. We demonstrated that it is possible to extract meaningful insights from these complex models. Our findings reveal that MGraphDTA learns to identify chemically relevant functional groups, rather than relying on superficial data patterns. The model's latent space exhibits a bias towards ligand features, reflecting the dataset's composition, and applies context-dependent reasoning through distinct "rule families" for different protein clusters. By generating specific, testable hypotheses about atom-residue interactions, this work represents a step towards building more trustworthy and interpretable deep learning models for drug discovery, transforming them into collaborative tools for scientific exploration.

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APPENDIX

A. Atom-to-Residue Interaction Matrix Example

Figure 12 shows an example of an atom-residue interaction heatmap generated using Integrated Gradients for the drug-protein pair used as demonstration in the core sections.

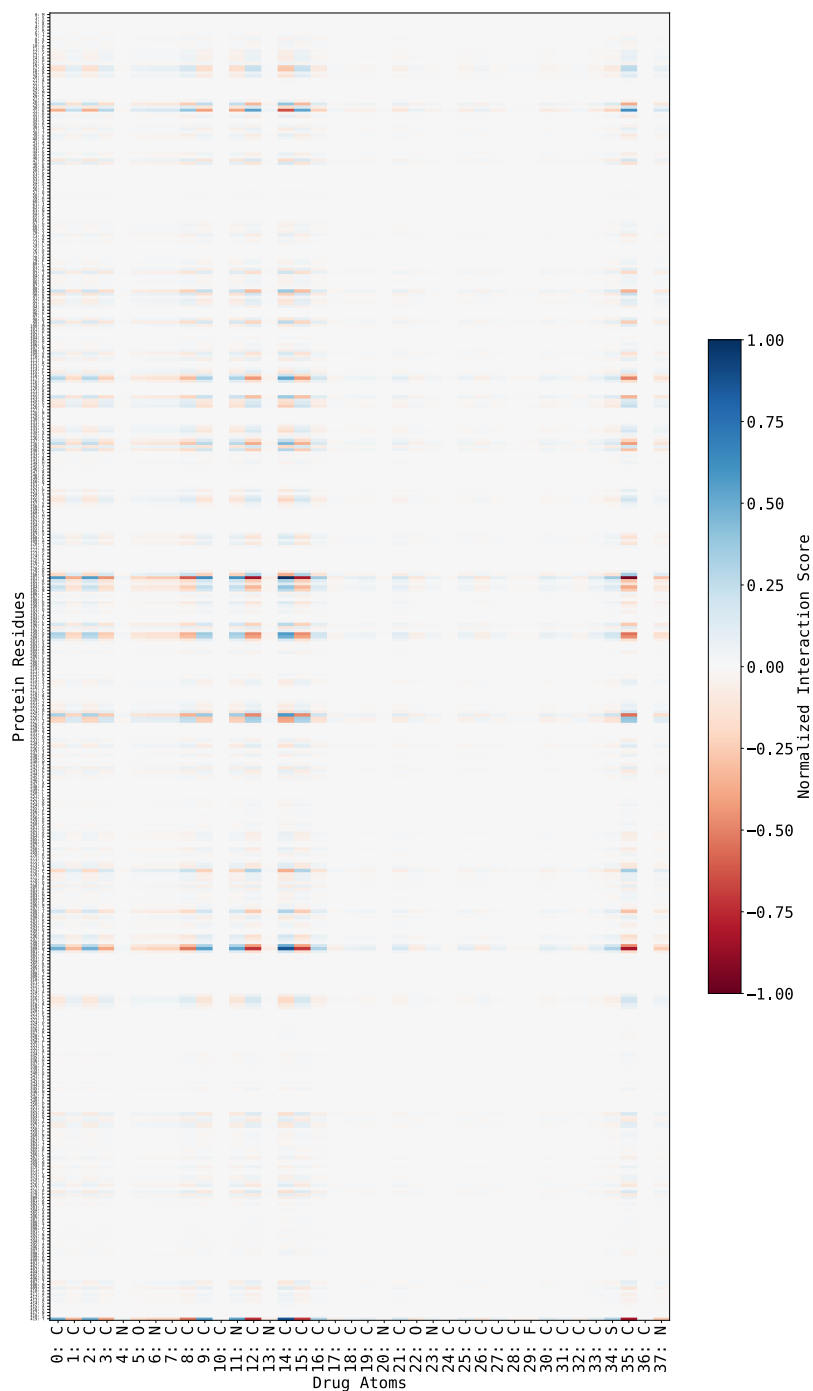
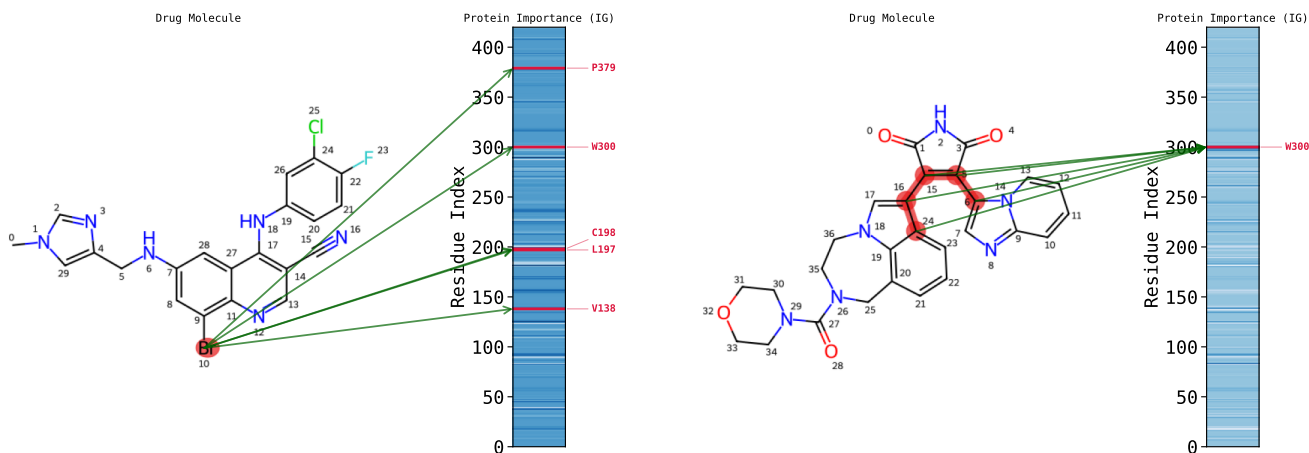


Fig. 12: Atom-residue interaction heatmap derived from Integrated Gradients. The x-axis represents drug atoms and the y-axis represents protein residues.

B. Additional Examples of drug-protein interaction visualizations

Figure 13 shows additional examples of drug-protein interaction visualizations generated using Integrated Gradients for a high-binding and a low-binding ligand. The drug is represented to the left, and the protein residues are represented in the bar to the right. The color intensity indicates the importance of each atom-residue interaction for the predicted binding affinity.

The protein is the same in both examples. Figure 13a shows the low-binding ligand for the protein, while Figure 13b shows the high-binding ligand for the same protein.



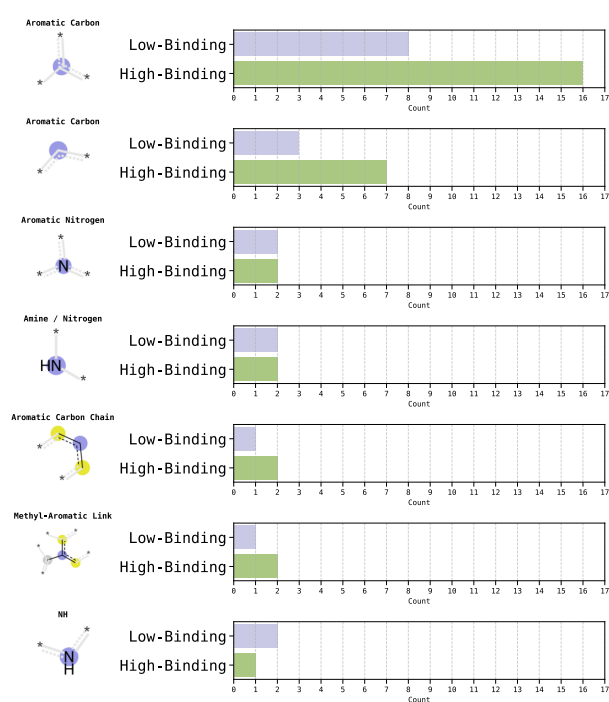
(a) Visualization of drug-protein interactions for a low-binding ligand using Integrated Gradients.

(b) Visualization of drug-protein interactions for a high-binding ligand using Integrated Gradients.

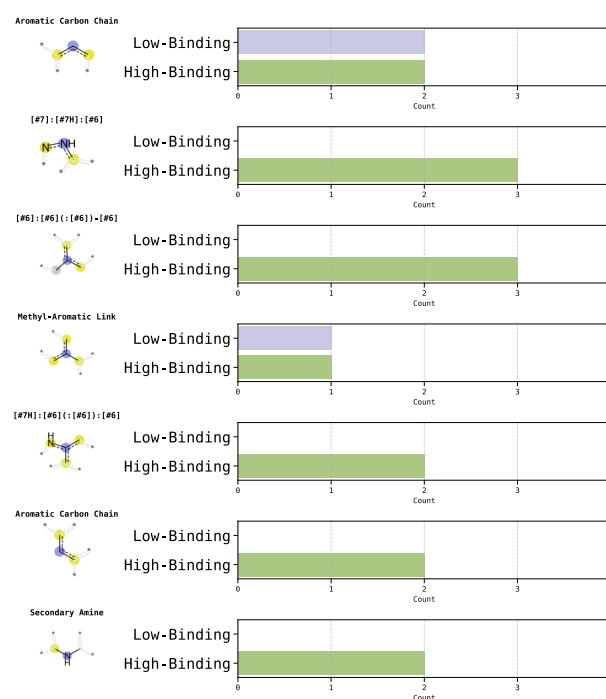
Fig. 13: Additional examples of drug-protein interaction visualizations generated using Integrated Gradients for (a) a low-binding ligand and (b) a high-binding ligand against the same protein target. The drug is represented to the left, and the protein residues are represented in the bar to the right. The color intensity indicates the importance of each atom-residue interaction for the predicted binding affinity.

C. Comparison of Important Chemical Substructures for High- vs. Low-Binding Ligands

Figure 14 presents a comparative analysis of the most important chemical substructures identified by GNNExplainer for the top-20 high-binding versus top-20 low-binding ligands tested against the same protein target.



(a) Comparative frequency of the most important chemical substructures for top-20 high- vs. top-20 low-binding ligands tested against the same test protein target. This visualization considers all atoms, including atom-only substructures.



(b) Comparative frequency of the most important chemical substructures for top-20 high- vs. top-20 low-binding ligands tested against the same test protein target. This visualization excludes atoms at zero radius, focusing on larger functional groups, providing extended context.

Fig. 14: Comparison of important chemical substructures identified by GNNExplainer for high-binding vs. low-binding ligands against the same protein target. (a) includes all atoms within the substructure, while (b) excludes atom-only substructures to focus on larger functional groups.